

40-year annual time series of dominant tree species and leaf type cover for Italy derived from Landsat using deep learning

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Abstract. Consistent long-term maps of forest functional type and dominant tree species cover at high spatial resolution are currently unavailable for Italy. The study presents annual 30 m resolution maps of continuous evergreen broad-leaved, deciduous, and coniferous leaf type cover fractions and dominant tree species class probabilities for all forested pixels in Italy spanning 1987–2025, derived from the Landsat Collection-2 Level-2 archive (Landsat 4/5/7/8/9). Two Time Series Transformer models are trained via a three-stage workflow combining contrastive pretraining with supervised fine-tuning on Italian forest plot observations, forest inventories, and a synthetically generated training dataset to predict both tasks. Pixel-level uncertainty estimates are provided as the standard deviation of a five-seed deep ensemble for both models. A three-year observation pooling strategy is applied to increase cloud-free observation density and mitigate inter-annual variability. Evaluation of the time series maps are based on hold-out areas and a comparison with historical orthophotos. The 39-year time series enables historical reconstruction of forest composition dynamics and long-term ecological change detection at national scale and relevant time scales for climate-driven species range shifts.

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1 Introduction

Landsat for long term forest monitoring: With nearly 40 years of continuous acquisitions, the Landsat archive constitutes the longest consistent Earth observation record available for global vegetation monitoring (Wulder et al., 2022). Its four decades of continuity makes it uniquely suited for change detection or hindcasting of environmental variables including forest habitat characteristics (Bell et al., 2024; Turubanova et al., 2023). The archive has been applied to discriminate forest types and quantify forest variables such as canopy height or greening trends from regional to conti-

mental scales (Pflugmacher et al., 2019; Obuchowicz et al., 2024; Yel et al., 2026). In monitoring long-term changes and disturbances in forest vegetation under climate change, the Landsat archive provides a valuable modelling base (Viana-Soto and Senf, 2025; Grünig et al., 2026). The relatively slow change and succession dynamics of forest vegetation, which is unfolding over decades, make this long temporal availability particularly valuable. A shorter observation period would fail to capture natural stand-level transitions such as species shifts or forest densification, which happen gradually over long time windows (Tong et al., 2023; Midolo et al., 2026). The medium spectral, spatial, and temporal resolution of Landsat (6 reflectance bands, 30 m ground sampling distance, and 8–16 day revisit time) is adequate for capturing

annual phenological dynamics at stand level, which demonstrated to be critical for distinguishing tree species and functional leaf types (Kollert et al., 2021; Hemmerling et al., 2021; Wang et al., 2022).

Challenges: Despite these strengths, the use of multi-decadal Landsat data for vegetation mapping raises several methodological challenges. The number of usable cloud-free observations has increased since the early periods of the archive, due to the addition of new Landsat satellites. This systematic increase in observation density has been shown to cause spurious trends in annual maximum NDVI, where up to 50% of observed Landsat-based greening in alpine environments can be attributed to observational sampling bias rather than true vegetation change (Bayle et al., 2024). A second challenge arises from the failure of the Scan Line Corrector (SLC-off) on Landsat 7 ETM+, which introduces systematic spatial data gaps. Finally, spectral radiometric differences between sensors — arising from different sensors, band widths, and data processing procedures across Landsat missions — introduce inter-sensor discontinuities that may produce spurious changes in modelled variables across the 40-year record.

ML/DL for dominant tree species and leaf type cover mapping: Multitemporal satellite data consistently outperform single-date or mosaic-based approaches for tree species and leaf type mapping. Phenologically informative seasons — particularly leaf-off periods and autumn senescence — contribute disproportionately to species discrimination (Kollert et al., 2021; Tan et al., 2025). When sufficient temporal coverage is available, dense time series exploiting the full phenological cycle can outperform the inclusion of additional environmental predictor variables such as topography or climate indices (Hemmerling et al., 2021). These findings highlight temporal resolution and coverage as primary sources for improving species-level modelling and mapping performances.

Deep learning, and in particular Transformer-based architectures, provide a framework for processing raw, irregular, multi-band satellite time series (STS). Transformers process sequences of observations without requiring interpolation to a regular sequence, an essential property for long Landsat archives with varying observation density across decades and sensor platforms (Zerveas et al., 2020; Yuan et al., 2022). Self-supervised pretraining on large unlabelled STS or image time series datasets has been shown to improve representation quality and performance on downstream tasks, particularly in data-scarce settings (Yuan Yuan et al., 2020; Hiebl et al., 2025; Feng et al., 2026). Contrastive learning approaches exploit temporal and seasonal variability by training representations to be invariant across different observation windows of the same location — a property suited to stable land cover types such as forest stands (Mañas et al., 2021; Chen et al., 2020). Providing calibrated uncertainty estimates alongside predictions has emerged as an important requirement for ecological mapping applications, to enhance

interpretability by users. Deep ensembles of independently trained networks deliver calibrated epistemic uncertainty that reliably identifies out-of-distribution inputs and unreliable predictions, when feature and label related domain shifts appear during the mapping phase (Sylvain et al., 2024; Hiebl et al., 2025).

Relevance for Italys forests: Italian forests are undergoing rapid and spatially complex changes. The Italian National Forest Inventory (INFC2015) documents a continuing expansion of forested area from approximately 10.46 million ha in INFC2005 to 10.98 million ha in INFC2015 driven by land abandonment and targeted reforestation programmes (Gasparini et al., 2022). At the same time, land abandonment in combination with climate change alters phenological characteristics and forest composition in favor of species with higher drought resilience (Herraiz et al., 2025; Chelli et al., 2017), while evergreen broad-leaved species are spreading in deciduous forest areas (Conedera et al., 2018). These processes are spatially heterogeneous, reflecting regional variety in land use changes, climate change, and historical species composition. Despite these dynamics, spatially explicit maps of forest composition and variables in Italy remain restricted to static maps with no continuous temporal resolution. The most recent national reference product, the Carta Forestale d'Italia 2020 (CFI2020), provides a single-epoch wall-to-wall forest map at 1:10,000 scale based on photo-interpretation of aerial orthophotos, but does not capture interannual dynamics or historical trajectories of change (Mattioli et al., 2025).

What will be done? Our study will fill this gap by providing data-driven, spatio-temporal continuous maps of leaf type cover (LTC) fractions and dominant tree species (DTS) probabilities for all forested areas in Italy - annually and at Landsat-native 30m resolution. We assemble and process 40-years of Landsat Collection-2 Level-2 time series from 1985 to 2025 (Landsat-4/5/7/8/9) and train multiple Time Series Transformer models (TSTpad) via contrastive pretraining and a fine-tuning workflow. The final product are annual 30m maps of i) 11 dominant tree species classes, and ii) LTC fractions for evergreen broad-leaved (EVE), deciduous broad-leaved (DEC) and coniferous (CON) functional leaf type. Additional ensemble-based uncertainty estimations will be provided per-pixel and target variable.

2 Materials and Methods

2.1 Study Area

The study covers all forested areas of peninsular and insular Italy, spanning several biogeographical, climatic, and latitudinal regions and the full elevation gradient of the country (Fig. 1). Forest extends from sea level — 34.1% of the national forest area lies below 500m a.s.l. and 1.1% above 2000m a.s.l. (Gasparini et al., 2022) — up to the alpine treeline, which in the western Italian Alps was dated at

2505 m a.s.l. in 2000 and 2515 m a.s.l. in 2008 (Leonelli et al., 2011). Italian forests fall within alpine, temperate, sub-Mediterranean, and Mediterranean climate zones (Chelli et al., 2017), and are dominated by four forest habitat types: broadleaved-deciduous and temperate coniferous forests concentrated in the Alps and Apennines, and evergreen broad-leaved and Mediterranean coniferous forests concentrated along the coasts and in the south (Bricca et al., 2026). This climatic and ecological heterogeneity is a central challenge for wall-to-wall mapping, as models trained predominantly on well-sampled regions risk out-of-domain predictions in climatically and ecologically different areas with sparser reference data. Forests cover 37% of Italy's territory (10.98 million ha) (Gasparini et al., 2022). Italy's position as a transition zone between the Mediterranean and the temperate climate of Central Europe makes its forests especially vulnerable to climate change: warming and increasing drought frequency are already documented to shift phenology and favour more drought-resilient and evergreen broad-leaved species at the cost of deciduous ones (Chelli et al., 2017; Herraiz et al., 2025; Conedera et al., 2018). These climate-driven shifts compound ongoing changes in forest management and land use, jointly reshaping tree species distribution and biodiversity across the country (Bricca et al., 2026).

2.2 Modelling Data

An overview of the used datasets during training are provided in Tab.1 and Fig.2.

2.2.1 Italian forest vegetation plot databases

The Forest Vegetation Database Italy (VDBI) is a compilation of over 16,000 phytosociological and vegetation relevé plot observations collected across Italy from the 1970s to the present. Each plot record contains species composition data, structural attributes, and cover estimates for the tree strata (Alessi et al., 2019). The VDBI covers the full range of Italian forest types from Mediterranean macchia to sub-alpine conifer belts. It provides temporal depth extending back to approximately 1972, enabling historical training data for pre-2000 Landsat time steps. The location accuracy of single plot observations in the dataset varies across observation year and source dataset. To avoid non-forest pixels leak into the training and test datasets, VDBI was cleaned manually based on orthophoto imagery. The final dataset after the cleaning process provides 9837 plot observations. Plot records include LTC estimates (EVE, DEC, CON) and tree species cover fractions, which were used to retrieve dominant DTS classification labels, allowing joint training of the regression ($VDBI_{DTS}$) and classification ($VDBI_{DTS}$) TST-pad model. A 10×10 km regular polygon grid was created for the train/validation/test split to reduce spatial autocorrelation between training and validation sets. A 20% validation

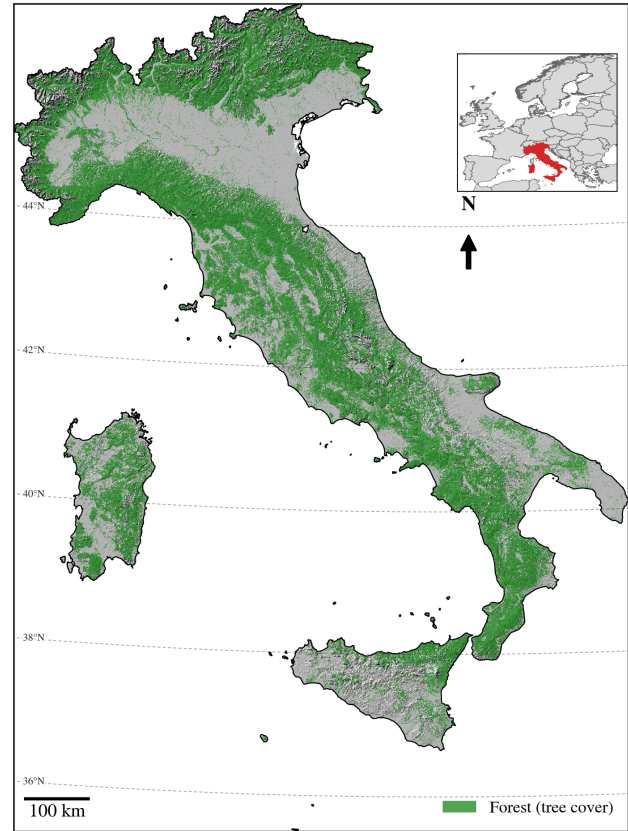


Figure 1. Study Area Italy with high forest cover in the Alps and mountainous areas of Central Italy.

split, stratified by polygon identifier is applied throughout all training stages (Safonova et al., 2023).

2.2.2 Field plot observations

The Vegetation Plot Observations (VPO) dataset consists of recent field campaigns conducted within 7 Italian National Parks or protected areas (see: Hiebl et al. (2025, 2026)). Each site comprises four systematic 100 m² vegetation plots arranged on a regular grid, within which species-wise cover is estimated for three vertical layers (0–1 m, 1–10 m, and above 10 m). LTC is extracted as sum of EVE, DEC, and CON species cover fractions, while DTS is - similar to VDBI - retrieved from species cover by selecting the tree species with highest cover fraction. The VPO data used in this study consists of observations from 2023 to 2025 field season, yielding a total of 2016 individual plot observations from 504 sites. The VPO dataset provides high-quality, spatially clustered reference data for the most recent Landsat time steps (post-2020) and serves as the primary validation reference for model performance in the recent high-density observation period. The VPO dataset is further split into a training set (80%) and a validation set (20%) to evaluate model performance. The split was performed spatially after clustering

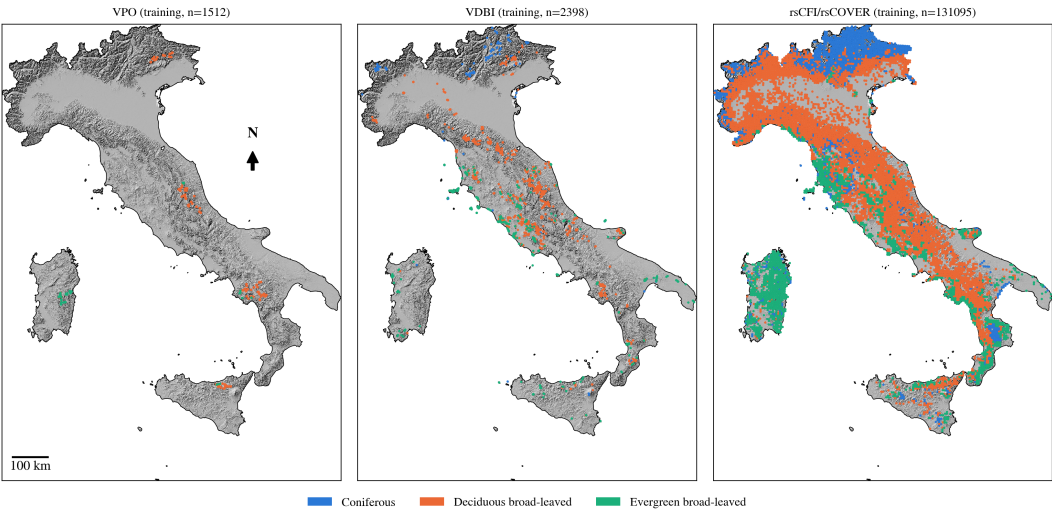


Figure 2. Overview of training datasets used in the study.

Table 1. Overview of datasets used for model training and validation.

	n samples	LTC	DTS	
VPO	2016	yes	yes	conducted in the field 2023–2025, standardized 10 × 10 m plots (see Sec. 2.2.2)
VDBI	2998	yes	yes	observations with varying size, acquisition time, and location accuracy (0.01 – 10 km) (see Sec. 2.2.1)
rsCFI	131095	-	yes	randomly sampled across CFI dominant tree species (see Sec. 2.2.3)
csCFI	30380	-	yes	pure stand polygons sampled from CFI (see Sec. 2.2.3)
rsCOVER	131095	yes	-	same locations as rsCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4)
csCOVER	30380	yes	-	same locations as csCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4)

the sites by their coordinates into 10 clusters per region. The spatial clustering was done to stabilize training and prevent over-optimistic accuracy estimates due to spatial autocorrelation (Safonova et al., 2023; Hiebl et al., 2025).

2.2.3 CFI Forest Inventory Data

The CFI2020 is the first national forest map of Italy produced at operational scale (1:10,000), based on photointerpretation of aerial orthophotos from 2018 to 2020 against three simultaneously applied forest definitions (Mattioli et al., 2025). The CFI2020 delineates forested areas across all 21 Italian regions and provides stand-level attributes including forest category, silvicultural system, and DTS group. The initial 43 unique classes in the dataset were collapsed into 11 dominant tree species classes: silver fir, spruce, pine, deciduous oak, chestnut, beech, manna ash - hornbeam, evergreen oak, coniferous others, deciduous others, evergreen others. In this study, the CFI2020 polygon layer serves two roles: i) as an additional training/validation data source for the DTS classification task, and ii) as a source of decision rules for cleaning the synthetic LTC training dataset (see 2.2.4). In the first case two different datasets are created by i) randomly sampling a

set of evenly distributed points over all classes (rsCFI), and ii) by sampling points from manually delineated pure stand polygons that are evenly distributed across classes (csCFI). In the second case, CFI2020 species attribution (DTS) is used to impose ecologically plausible constraints on predicted cover fractions of the synthetic dataset (see 2.2.4).

2.2.4 Synthetic leaf type cover data

Spatially explicit wall-to-wall leaf type cover data for model training are not available for Italy. Following the approach of Kang et al. (2021) and Tan et al. (2025) — who demonstrate the use of coarser or indirectly derived data sources as proxy training targets where direct measurements are absent — an artificial leaf type cover dataset was created from the VDBI and VPO plot observations. A Random Forest regression model was trained on the combined VDBI and VPO training split, with EVE, DEC, and CON cover fractions as prediction targets. As input features, AlphaEarth embeddings (Brown et al., 2025; Hiebl et al., 2026) — multimodal spectral-spatial-temporal feature representations derived from a globally pretrained geospatial foundation model — were extracted for all training plot locations. AlphaEarth

embeddings were shown to achieve high species classification accuracy with very limited training data in Italian mountain forests (Ball et al., 2026), making them particularly suited as inputs for the pseudo-label generation step.

5 The trained Random Forest model was then applied to the full CFI2020 forest extent to generate a nation-wide pixel-level pseudo-label layer for each cover fraction. The labels were extracted at the same locations as rsCFI (rsCOVER) and csCFI (csCOVER). This pseudo-label layer constitutes the
10 primary continuous-cover target during the contrastive pre-training and early fine-tuning stages. To reduce label noise introduced by model prediction errors, the CFI2020 species attribution was used to apply ecologically motivated cleaning rules: e.g. pixels within pine-dominated forest polygons
15 were required to have at least 60% CON cover and fewer than 30% EVE cover; analogous rules were applied to classes of beech, oak, and dominant tree species. These cleaning rules remove the most pseudo-label errors while preserving the spatial completeness of the training dataset. If a sampled syn-
20 thetic plot does not meet the criteria the sample is dropped. This way we prevent ecologically non-feasible labels being forwarded into the final training loop.

2.2.5 Landsat time series

figure: landsat time series plot over all years for 2 different
25 dominant tree species, include modelling approach with 3 year window aggregation

Annual Landsat time series were assembled for the full Italian national territory from Microsoft Planetary Computers STAC Catalog using Landsat Collection-2 Level-2 surface reflectance products from Landsat 4 TM (1982–1993),
30 Landsat 5 TM (1984–2013), Landsat 7 ETM+ (1999–2023), Landsat 8 OLI (2013–present), and Landsat 9 OLI-2 (2021–present). Scenes with cloud cover higher than 75% were excluded, which led to a per year count of approx.
35 80 to 120 scenes. Per-pixel cloud, cloud shadow, cirrus, and snow masking was performed using the Landsat Collection-2 Level-2 QA PIXEL quality bitfield. Per-observation quality flags derived from QA PIXEL are propagated through the entire modelling pipeline as a key-padding mask, ensuring
40 that cloud-contaminated or padded time steps are ignored by all attention layers of the TSTpad model. For each forested pixel, 14 spectral features are computed at each valid observation: six surface reflectance bands (blue, green, red, near-infrared, SWIR1, SWIR2) and eight derived spectral indices
45 (NDVI, NIRv, NDMI, EVI, WDRVI, NBR, NDWI, GNDVI). At the training plot location the resulting multi-annual time series are stored as Zarr-compressed arrays to enable efficient random-access loading during model training. For mapping, the Landsat scenes were fetched and preprocessed on-the-
50 fly in 25×25 km chunks from Microsoft Planetary Computer parallel to the GPU-based model prediction.

2.2.6 Additional data sources

The ESA WorldCover product is used as an initial broad-scale delineation of forested areas across Italy (Zanaga et al., 2022). Only pixels labelled as tree cover are forwarded for
55 prediction in the final maps. We are aware that the ESA WorldCover product does not provide a sufficient forest mask for detailed analysis of the datasets, as many mixed pixels and edge cases are remaining. But by applying a conservative masking during mapping we leave room for stricter forest
60 delineation through other products without loosing regional accuracy.

2.3 Time Series Transformer Architecture

The backbone model is an encoder-only Time Series Transformer extended to support irregular, sparse, annual satellite
65 image time series (TSTpad). The architecture employs multi-head self-attention across the temporal dimension, enabling the model to learn both short-range phenological features and long-range inter-seasonal dependencies within the aggregated 3-year observation window (Rußwurm and Körner,
70 2020; Zerveas et al., 2020).

For each target year, three consecutive years of Landsat observations are concatenated, encompassing the target year and the two immediately preceding years (year-2, year-1,
75 year), and aggregated into a single annual cycle to densify the sparse annual time series. After aggregation, observations are chronologically sorted and padded to a fixed sequence length of 200 time steps. Positions corresponding to padded time steps receive a key-padding mask that prevents these
80 positions from contributing to attention score computations (Vaswani et al., 2023). The actual acquisition day-of-year for each observation is encoded as a sinusoidal positional encoding, enabling the model to correctly represent the irregular temporal spacing of Landsat acquisitions and to distinguish
85 phenological phases across the observation window without requiring interpolation to a regular grid (Yuan Yuan et al., 2020). Input features are normalised per band using a robust standardisation based on the median and interquartile range (quantiles $q=0.1$ and $q=0.9$) — with transform parameters fitted on the training data and frozen at inference time
90 to prevent data leakage. A sensor platform embedding with $\text{sensordim} = 5$ encodes a multi-hot vector per year indicating which Landsat satellites (L4, L5, L7, L8, L9) contributed to the observations. This embedding is added to the pre-attention temporal encoding via a learned linear projection
95 and makes the Transformer aware of inter-sensor radiometric and observation density differences over the 40-year archive. A post-attention pooling layer using a learned query vector collapses the temporal dimension into one single summary vector per output target, letting each targets learned query de-
100 cide, which valid timesteps matter most for its prediction. In a setting with target classes from a wide range of phenological and spectral characteristics, this pooling strategy allows

the model to learn class-specific temporal patterns across the observation window. The model heads are two-layer Multi-Layer-Perceptrons including Dropout and GeLU activation functions (Hendrycks and Gimpel, 2016). The classification model heads map the Transformer’s output to class probabilities over eleven dominant tree species and three functional leaf type classes trained with two separate Cross Entropy Loss (CELoss) computations summed into a single loss value (dominant species and functional leaf type). With the dual loss the model learns to simultaneously capture species identity without losing the ability to capture the distinction between EVE and CON leaf types. The regression heads trained with Mean Squared Error Loss (MSELoss) maps the output to continuous EVE, DEC, and CON cover fractions.

2.4 Training procedure

TODO workflow figure.

The model is trained via a three-stage progressive workflow designed to maximise temporal stability and to accommodate the heterogeneous quality and temporal coverage of available reference data.

Stage 1 - Contrastive pretraining. The backbone is trained using a joint loss that simultaneously optimises a supervised classification objective (DTS or LTC) using CELoss/MSELoss and an unsupervised normalised temperature-scaled cross-entropy contrastive loss (NT-Xent) (Chen et al., 2020): $L = \lambda \cdot \text{NT-Xent} + \text{CE}$. For the contrastive term, positive pairs are formed from the same forest plot observed in two independently sampled three-year windows ($year_0$ and $year_1$ drawn at random), representing different but ecologically equivalent observation contexts for the same location. NT-Xent pushes the backbone representations of these positive pairs toward each other in embedding space while repelling representations of observations from different plots. This objective forces the backbone to learn year-invariant spectral-temporal features that are not confounded by observation density or inter-annual radiometric variability (Mañas et al., 2021). Positive pairs are further diversified by time series augmentation, which applies three independent perturbations to each view: random temporal masking (up to 20% of valid time steps zeroed), spectral noise injection (additive Gaussian noise with $\sigma = 0.01$), and band masking (10% of spectral bands zeroed per sample) (Yuan et al., 2025). For contrastive pretraining only data from csCFI and csCOVER are used, as these pure forest stands are considered stable over time. The contrastive and classification signals are routed through separate heads attached to the shared backbone: a two-layer MLP projection head for NT-Xent, and the classification/regression head for CELoss or MSELoss, respectively. The model is optimised using AdamW with a cosine learning rate schedule preceded by a 10% warm-up phase. Stage 1 training runs for 60 epochs with a batch size of 128. Reference data consist of csCFI, rsCFI, csCOVER, and rsCOVER. Class imbalance is addressed jointly by class-

weighted CrossEntropyLoss and by sample weights set to the square root of the inverse class frequency, following the approach of Grabska-Szwagrzyk et al. (2024). A stratified 20% validation hold-out, stratified by plot polygon identifier, is maintained across all training stages to prevent spatial autocorrelation between training and validation sets.

Stage 2. Fine-tuning on labelled data. The contrastive backbone from Stage 1 is loaded and fine-tuned under a purely supervised objective (CrossEntropyLoss + MSELoss). The embedding layer is kept frozen in this stage to preserve the contrastive representations, while the task heads are retrained using a reduced learning rate to prevent catastrophic forgetting (Hiebl et al., 2025). This stage runs for 10 epochs on the full 1985–2025 Landsat time series, allowing the model to adapt the task heads to the full historical archive without disturbing the temporally invariant representations established during contrastive pretraining. The used reference data are the VDBI historical plot observations.

Stage 3. Fine-tuning on plot observation data. Additional fine-tuning passes are applied from the Stage 2 checkpoint giving priority to recent high-quality VPO field observations that provide the most accurate ground reference data for the present-day forest state. This stage is particularly intended to anchor the model’s predictions of recent (post-2020) DTS and LTC to field-validated reference values, while the contrastive representations established in earlier stages ensure that this anchoring remains consistent with historical predictions.

Five independent model instances, differing in random seed, architecture micro-choices within TSTpad, and random subsets of training and validation data, are trained per stage. The five-seed ensemble delivers two outputs per pixel per year: the ensemble mean prediction (used as the primary data product) and the inter-seed standard deviation (used as the pixel-level uncertainty estimate). High inter-seed uncertainty identifies pixels where the model predictions are unstable, typically in regions of low observation density, unusual spectral composition, or high class confusion.

2.5 Mapping scheme

The forest prediction domain is defined by the ESA Word-Cover forested area (see 2.2.6). For each target year from 1985 to 2025, the corresponding three-year Landsat observation window (year-2 to year, inclusive) is assembled from the Landsat Collection-2 Level-2 archive, cloud-masked, and processed to the 14 spectral features described in Section 2.2.5. Features are normalised with the fixed training-time standardization parameters and padded to 100 time steps. The five trained ensemble models are applied independently to each forested pixel, and the resulting class probability distributions and cover fraction predictions are aggregated across ensemble members to produce the mean prediction and inter-member uncertainty layers.

Observation density varies substantially across the archive. Before 1999, when only Landsat 5 TM was operational, the three-year window typically yields fewer cloud-free observations than the post-2013 period when Landsat 8 and 9 operate simultaneously. This density gradient is an analogue of the observational sampling bias documented for NDVI trend analyses by Bayle et al. (2024): predictions in the early archive period carry higher uncertainty due to sparse temporal sampling, which may cause the model to have lower confidence in phenological-season attribution.

We did not process the resulting annual time series using outlier detection or temporal smoothing, before storing the published LTC and DTS prediction maps. This was a design choice so downstream users can - and are encouraged to - apply temporal smoothing or outlier detection methods specifically appropriate to their application context. This is a necessary step for downstream tasks, as the time series contain both true ecological change, but also spurious noise from observation gaps, cloud contamination, and inter-annual radiometric variability.

2.6 Evaluation

The validation strategy is designed to assess both the accuracy of the map product at recent time steps and the temporal consistency of predictions across the full 40-year archive. Three independent validation datasets are employed. The VDBI hold-out test split provides the primary historical validation reference, with plot observations dating back to approximately 1995. This split allows accuracy assessment at historical time steps (1995–2025) where field data are available, spanning the Landsat 5-only and Landsat 5/7 overlap periods. The VPO hold-out split — collected during the 2023 to 2025 field work — provides the primary validation reference for recent predictions (post-2020) from the high-density Landsat 8/9 period. This dataset contains data from many mixed forest stands with hard-to-predict labels. The two validation datasets provide complementary temporal coverage: the VDBI split quantifies historical accuracy under lower observation density, while the VPO split quantifies accuracy under the best available observation conditions. A third validation source consists of a set of pure-stand forest polygons manually interpreted from high-resolution Google Earth Pro imagery in combination with the CFI dataset at recent time steps. These polygons, selected to represent spectrally unambiguous mono-specific stands, enable validation of the classification output under conditions where the expected dominant species label is highly reliable, providing an upper bound on classification accuracy unconstrained by label uncertainty in mixed stands or uncertain plot observation coordinates.

For the classification head, overall accuracy (OA) and per-class F1 score are computed for each period. For the regression heads, root mean squared error (RMSE) and mean absolute error (MAE) are reported for each of the three cover

fractions (EVE, DEC, CON). The VPO dataset is finally used to assess in detail the model's ability to predict DTS and LTC in mixed stands, where the prediction of a single label is inherently more difficult and uncertain.

To account for uncertainties in plot observations coming from (i) a spatial mismatch between plot location and Landsat pixel, and (ii) temporal uncertainty and variations between plot observation year and Landsat acquisition year, we complement the direct accuracy metric with a true-label-conditioned metric (Lobo et al., 2020). Predictions are evaluated over a space-time neighborhood around each plots reported coordinates and year, weighting candidate locations by a Gaussian space-time prior (location and temporal uncertainty) combined with a Gaussian likelihood that down-weights candidates whose predicted value strongly disagrees with the plots true label, following the logic of likelihood-based geolocation correction (Shannon et al., 2024). This estimator is bracketed by two limiting cases obtained by varying the likelihood bandwidth: a pure prior-weighted neighborhood mean that ignores the label entirely, and a best-match estimate that selects the single most consistent candidate location. We report accuracy metrics (RMSE, MAE and R^2) computed under all three methods to characterize how sensitive model accuracy is to possible location and temporal errors (see Appendix A).

3 Results and Discussion

The final maps provide following variables per pixel and per year (1987 to 2025):

- Dominant tree species: ensemble mean of dominant tree species class and level of agreement across ensemble members; ensemble mean of class probabilities and ensemble standard deviation for 11 classes
- Leaf type cover fractions: ensemble mean of EVE, DEC, and CON cover fractions and ensemble standard deviation

3.1 Performance metrics on test datasets

3.1.1 Leaf type cover regression

The hold-out splits of the recent, methodologically homogeneous VPO dataset and of the multi-temporal VDBI collection are used to quantify the error of the annual LTC maps (Tab. 3). VPO plots were located with an accuracy that is high relative to the 30m Landsat grid, so the estimator evaluated directly at the reported plot coordinate is the appropriate headline figure for this dataset; the importance-weighted and best-match estimators are reported alongside it to characterise the sensitivity of the metrics to residual location and timing error (Sec. 2.6).

Evaluated directly at plot location, the maps reproduce DEC cover with an MAE of 9.7%_{cover} (RMSE 12.4%_{cover},

$R^2 = 0.86$) and EVE cover with an MAE of $12.5\%_{cover}$ (RMSE $19.1\%_{cover}$, $R^2 = 0.66$). The consistently weaker performance for EVE has two plausible and non-exclusive explanations. First, the ratio of RMSE to MAE is larger for EVE ($6.6\%_{cover}$) than for DEC ($2.7\%_{cover}$), indicating that the EVE error is driven by a comparatively small number of large residuals rather than by a uniform offset, i.e. by a subset of stands that is systematically misinterpreted rather than by generally noisier predictions. Second, VPO reference cover is recorded separately for three vertical strata and aggregated across them, whereas Landsat observes an integrated top-of-canopy signal. Evergreen broad-leaved species in Italy frequently occur as a sub-canopy or understory component beneath a deciduous overstory, a configuration that is only weakly expressed in canopy reflectance outside the leaf-off window (Chastain and Townsend, 2007). Reference EVE cover can therefore exceed the fraction of the signal that is physically available to the sensor, which inflates the apparent model error without a corresponding error in the mapped canopy composition.

CON cover attains the lowest absolute errors on the VPO split (MAE $5.0\%_{cover}$, RMSE $9.1\%_{cover}$) but a negative coefficient of determination ($R^2 = -0.05$). This combination is a property of the evaluation sample rather than of the model: the VPO campaigns contain almost no coniferous-dominated stands, so the reference CON variance against which R^2 is normalised is very small, and a near-constant prediction is penalised even when its absolute deviation is minor. We therefore do not interpret R^2 for CON on the VPO split, and CON accuracy in the recent period should be judged from the CON-bearing subsets of the VDBI and CFI evaluations instead.

On the VDBI split the errors increase substantially for all three fractions (TODO: insert VDBI MAE/RMSE/ R^2 values once available and state the magnitude of the increase explicitly). This dataset aggregates relevés collected over five decades by different observers, many of them without or with low-accuracy GNSS positioning, with reported location accuracies ranging from 0.01 to 1 km (Sec. 2.2.1). Reference errors of this magnitude propagate directly into the accuracy figures: field reference error is known to bias remote sensing accuracy assessments pessimistically, and the effect becomes proportionally more pronounced the more accurate the predictions themselves are (Persson et al., 2022). Even under closed conifer canopy, recreational-grade GNSS positioning commonly deviates by several metres to tens of metres (Strunk et al., 2025), so a reported VDBI coordinate may address a Landsat pixel that is spectrally unrelated to the described stand. The VDBI figures are consequently not a clean estimate of model error, and we report them as a conservative lower bound on historical map quality rather than as an accuracy statement.

The three estimator variants bracket this effect quantitatively. For EVE on the VPO split, MAE decreases from 12.5% to 11.6% and further to $8.5\%_{cover}$, and R^2 increases from 0.66

to 0.72 and 0.79 , when moving from the direct estimator through the importance-weighted posterior to the best-match candidate. The gap between the direct and the best-match estimate is an upper bound on the fraction of the reported error that can be attributed to space-time reference mismatch rather than to the model, but it must not be read as a corrected accuracy value: the best-match estimator selects the neighbourhood candidate that agrees most closely with the reference label and is therefore optimistically biased by construction, in the same way that label-informed geolocation correction requires an independent high-accuracy reference to be interpretable as unbiased (Shannon et al., 2024). The true accuracy lies between the direct and the best-match figures, and we use the direct estimate as the reference value throughout.

3.1.2 Dominant tree species classification

Classification performance is assessed on the VPO, VDBI, and CFI test datasets (Tab. 2). The model reaches a weighted-average F1-score of 0.62 on the VPO split (macro F1 0.65 over the six classes represented), 0.64 on the VDBI split (macro F1 0.60), and 0.72 on the CFI test dataset (macro F1 0.71 over all eleven classes). (TODO: report overall accuracy per dataset once the corresponding cells in Tab. 2 are filled.)

The per-class results are markedly asymmetric between precision and recall, and this asymmetry is more informative than the aggregate scores. On the VPO split beech is recovered with high recall but low precision ($0.91 / 0.57$), whereas deciduous oak shows the inverse pattern ($0.46 / 0.88$): the model assigns beech to a substantial share of oak-dominated plots, but the beech plots it does identify are largely correct. The same directional confusion appears in the VDBI and CFI evaluations, and it is consistent with the well-documented mutual confusion of spectrally and phenologically similar temperate broadleaves reported for national-scale Sentinel-2 mapping (Grabska-Szwagrzyk et al., 2024; Hemmerling et al., 2021). The manna ash–hornbeam class (*Ornus–Ostrya*) is the weakest class across all three datasets (F1 0.43 – 0.63); these stands are typically structurally heterogeneous and successional, so a single dominant-species label is a poor description of the pixel in the first place. The aggregate “other deciduous”, “other coniferous” and “other evergreen” classes behave as expected for catch-all categories: on the VDBI split, “other deciduous” reaches a recall of 0.81 at a precision of 0.23 , i.e. it absorbs plots that the model cannot attribute to any specific class. Their scores should be read as a measure of residual class confusion, not as evidence that the underlying species are mapped.

Per-class figures based on small test supports require particular caution. The highest F1-score in the VPO evaluation (chestnut, 0.92) rests on 26 test plots, and the spruce entry in the VDBI evaluation on 15 plots; at these sample sizes the confidence intervals are wide enough that the values cannot be interpreted as class-level accuracy statements.

The difficulty of estimating reliable per-class accuracies for infrequent classes under strong class imbalance is a recurring limitation of large-area species mapping (Hemmerling et al., 2021; Grabska-Szwagrzyk et al., 2024). Five of the eleven classes — silver fir, spruce, pine, other coniferous and other evergreen — are not represented in the VPO split at all, because the 2023–2025 campaigns targeted protected areas in central and southern Italy. VPO therefore constrains the recent-period accuracy of the broad-leaved classes only, and the alpine conifer classes are evaluated exclusively through VDBI and CFI.

3.1.3 Pure versus mixed stands

Accuracy differs systematically between spectrally unambiguous and compositionally mixed reference plots. In pure stands, in which one of EVE, DEC or CON approaches 100%_{cover}, the regression error is considerably lower than in mixed stands with comparable shares of two leaf types (TODO: insert the stratified MAE/RMSE values for the pure and mixed subsets). The CFI test dataset yields the highest and most balanced classification scores of the three evaluations, but this comparison is not one of equal difficulty, and it is not one of equal independence. CFI test labels originate from photo-interpretation of the same CFI2020 product from which the rsCFI and csCFI training samples were drawn, and the csCFI polygons were manually selected to represent mono-specific stands. The CFI figures therefore describe the internal consistency of the model with its dominant training source under near-ideal label conditions, and are best understood as an upper bound on achievable accuracy rather than as an estimate of operational map accuracy. The same caveat applies to comparable national products: pure-stand F1-scores of 72–97 % reported for the German national tree species map decreased by 4–14 percentage points once mixed stands were included in the assessment (Blickensdorfer et al., 2024), and reference polygons for the Polish national map were restricted to stands with a species share of at least 90 % (Grabska-Szwagrzyk et al., 2024). Assessments built on pure stands alone systematically overestimate the accuracy that users encounter in mixed forest landscapes. Because a large share of VPO plots are mixed, the VPO figures are the most representative of realistic mapping conditions among the three evaluations reported here, and are correspondingly the most conservative.

3.1.4 Comparison with published products and scope of the reported metrics

The LTC accuracies are broadly comparable to published fractional composition products, although the comparison is indicative rather than strict. Direct prediction of tree species proportions from Sentinel-2 with a spectral–spatial CNN in Belgium achieved an adjusted R^2 of 0.50 for the predicted proportions and an overall accuracy of 0.73 for the majority

species (Bolyn et al., 2022), and the pan-European deciduous–coniferous proportion product derived from Sentinel-2 and harmonised national forest inventory plots reports R^2 values of 0.52–0.73 with relative RMSE between 30 and 65 % across processing areas (Miettinen et al., 2025). The DEC results reported here ($R^2 = 0.86$) fall above and the EVE results ($R^2 = 0.66$) within that range. Several differences caution against reading this as a like-for-like improvement: the reference quantity differs (projected cover per leaf type in this study versus basal-area or biomass proportions in the cited products), the sensor and grid differ (30 m Landsat versus 10 m or 2.5 m Sentinel-2), the class definitions are coarser here, and the evaluation samples differ in spatial design.

This last point constrains the interpretation of all metrics reported in this section. The VPO, VDBI and CFI hold-out splits are spatially clustered, non-probability samples of the mapped domain: VPO is concentrated in a limited number of protected areas, VDBI reflects the historical sampling preferences of phytosociological surveys, and CFI samples were drawn to be evenly distributed across classes rather than across space. Spatial hold-out splitting at the polygon and cluster level was applied throughout to avoid the optimistic bias that arises when spatially autocorrelated observations are divided at random (Safonova et al., 2023), but this controls autocorrelation between training and test data without making the test sets representative of the mapped area. The reported values therefore characterise per-pixel model performance at reference locations and are explicitly not area-adjusted estimates of map accuracy in the sense of Olofsson et al. (2014). Users requiring area-unbiased accuracy or area statistics for a specific region or period should conduct a dedicated probability-based accuracy assessment on the published maps, and should consult the per-pixel ensemble standard deviation layers when delineating the domain over which the maps are used.

TODO: probabilistic metrics (calibration of class probabilities, reliability of the ensemble standard deviation) for all datasets.

TODO: confusion matrix figure for the DTS task – essential for showing which class pairs are confused and how far these coincide with typical mixed-stand species combinations.

3.2 Dominant tree species dynamics

TODO. - figure of transitions of 3 most likely transformations - figure of biggest areas of change - table per class of highest losses/gains

3.3 Leaf type cover dynamics

TODO. - figure of their sen slopes of the 3 target classes for all of Italy - table of forest type/habitat type with highest changes

Table 2. Classification performance (precision, recall, f1-score) for the dominant tree species classification task, evaluated on three test datasets. Reported is the likelihood-based importance-weighted performance result based on the location accuracy of the plots (see Sec. 2.6).

class	VPO test split				VDBI test split				CFI test dataset			
	prec.	recall	f1-score	support	prec.	recall	f1-score	support	pre.	recall	f1-score	support
silver fir	-	-	-	-	-	-	-	-	0.86	0.85	0.86	518
other dec.	0.52	0.55	0.53	112	0.23	0.81	0.36	27	0.37	0.85	0.52	240
dec. oak	0.88	0.46	0.60	127	0.71	0.40	0.51	123	0.67	0.61	0.64	1137
chestnut	0.96	0.88	0.92	26	0.89	0.62	0.73	30	0.67	0.74	0.70	817
other con.	-	-	-	-	-	-	-	-	0.49	0.50	0.49	220
beech	0.57	0.91	0.68	108	0.88	0.71	0.84	120	0.83	0.65	0.73	594
eve. oak	0.65	0.64	0.65	53	0.69	0.71	0.70	110	0.86	0.79	0.82	1141
other eve.	-	-	-	-	-	-	-	-	0.89	0.81	0.85	503
pine	-	-	-	-	0.90	0.82	0.86	51	0.74	0.72	0.73	410
ornus-ostr.	0.69	0.43	0.53	51	0.40	0.50	0.45	60	0.61	0.65	0.63	695
spruce	-	-	-	-	0.78	0.47	0.58	15	0.95	0.72	0.82	367
accuracy				493				536				30985
macro avg	0.71	0.65	0.65	493	0.67	0.61	0.60	536	0.72	0.72	0.71	30985
weighted avg	0.68	0.62	0.62	493	0.71	0.62	0.64	536	0.75	0.72	0.72	30985

Table 3. Regression performance (MAE, RMSE, R^2) for leaf type cover (LTC) prediction, evaluated on two test datasets. X/X/X refers to the computation of the evaluation metrics, being directly from plot location, importance-weighting, and best-match (see Sec. 2.6 and Appendix A). MAE and RMSE are reported as $\%_{cover}$.

target	VPO test split			VDBI test split		
	MAE	RMSE	R^2	MAE	RMSE	R^2
EVE	12.5 / 11.6 / 8.5	19.1 / 17.5 / 14.9	0.66 / 0.72 / 0.79	22.8 / 18.2 / 10.9	32.2 / 26.3 / 21.3	0.29 / 0.52 / 0.69
DEC	9.7 / 8.7 / 8.4	12.4 / 10.6 / 7.9	0.86 / 0.85 / 0.92	20.7 / 12.6 / 6.6	27.2 / 16.1 / 11.9	0.50 / 0.82 / 0.90
CON	5.0 / 4.4 / 3.0	9.1 / 7.4 / 6.1	-0.05 / 0.31 / 0.54	9.8 / 6.4 / 3.2	18.0 / 11.1 / 8.7	0.66 / 0.88 / 0.92

3.4 Limitations and future work

TODO.

- label noise and location accuracy issues (Persson et al., 2022; Strunk et al., 2025) in recent and historical plot observations. This affects the training massively, but also the test time performance reporting. Applying different computation techniques for metrics tries to tackle that but it does not compensate for reliable location measurements of independent test plots.
- Reliable multi-temporal reference datasets in forest mapping context are rare. The VDBI dataset is the only available multi-temporal dataset in Italy, but it is not representative for the whole country and has a high spatial bias.
- inter-annual spectral noise that leads to inconsistencies in per-pixel time series, assumption is a stable state or a single disturbance/gradual succession change of states. Random fluctuations can be considered noise. A Hidden Markov model + Vertibri smoother could help in smoothing the per-year argmax predictions and stabilize the probability transition vectors of each classes.
- need for a harmonized full scale Landsat archive as analysis

ready product (sensor calibration, sensor merge, density differences etc.)

- processing is huge limitation, time series into deep learning models (several weeks of mapping)
- foundation models for landsat time series (torchgeo) or/and embeddings -> easy/fast to use for processing long time series on large areas

4 Conclusions

The Landsat collections provide an unprecedented archive of satellite imagery for reconstructing and hindcasting environmental variables across large regions. Few studies exist, that exploit the full 40 years of available Collection-2 imagery to map forest related variables and provide time series datasets, that can be used to investigate forest dynamics and changes under climate and landuse change conditions. This study predicts annual maps of dominant leaf type and leaf type cover from Landsat time series, forest plot observations and forest inventories.

5 Code and data availability

All python code that was used during data processing, training and mapping is available under https://git.uibk.ac.at/c7161037/lis_mapping (training) and <https://git.uibk.ac.at/rslab/sattstools> (beta version, remote sensing data processing). The model architecture and model weights can be found on "link text". The Zarr-compressed annual maps of leaf type cover fractions and dominant tree species probabilities are available on "link text".

10 Appendix A: Computation of evaluation metrics

All accuracy figures reported in Sect. 2.6 are computed from the published maps, this means the estimators below are applied to the same spatio-temporal product that is published, and not to model outputs held in memory. For a plot p with reported coordinates s_p and observation year t_p , let $\mathcal{N}_p$ denote the set of candidate map pixels (s, t) with $\|s - s_p\|_\infty \leq 2\sigma_s$ and $|t - t_p| \leq 2\sigma_t$, clipped to the temporal extent of the product. This creates a squared box of half-width $2/\sigma_{\text{sigmas}}$ around the plot coordinates. The *direct* estimator is the special case $\mathcal{N}_p = \{(s_p, t_p)\}$, i.e. the nearest pixel of the reported year. The two estimators described below are conditioned on the reference plots label and are therefore validation diagnostics only, as they cannot be applied where the reference value is unknown. Pixels without a valid prediction receive zero weight and are excluded, and plot–prediction pairs containing a missing value are dropped before metrics are computed.

A1 Posterior-weighted estimator

Every candidate cell in $\mathcal{N}_p$ is treated as a hypothesis for the true plot location and observation date. The prior expresses the reported position and year and their uncertainty as a separable Gaussian kernel,

$$w^{\text{prior}}(s, t) = \exp\left(-\frac{\|s - s_p\|^2}{2\sigma_s^2}\right) \exp\left(-\frac{(t - t_p)^2}{2\sigma_t^2}\right), \quad (\text{A1})$$

with position uncertainty σ_s = plot location accuracy in m and temporal uncertainty $\sigma_t = 1$ a, clamped at $2\sigma_s$ and $2\sigma_t$. For the LTC task the likelihood of a hypothesis is the agreement between the mapped cover fraction $\hat{y}_k(s, t)$ of leaf type $k \in \{\text{EVE}, \text{DEC}, \text{CON}\}$ and the plot reference cover $y_{p,k}$,

$$w_k^{\text{lik}}(s, t) = \exp\left(-\frac{(\hat{y}_k(s, t) - y_{p,k})^2}{2\sigma_{\text{obs},k}^2}\right), \quad (\text{A2})$$

and the estimate is the posterior-weighted mean over the neighbourhood,

$$\hat{y}_{p,k}^{\text{PW}} = \frac{\sum_{(s,t) \in \mathcal{N}_p} w^{\text{prior}} w_k^{\text{lik}} \hat{y}_k(s, t)}{\sum_{(s,t) \in \mathcal{N}_p} w^{\text{prior}} w_k^{\text{lik}}}. \quad (\text{A3})$$

The observation bandwidth $\sigma_{\text{obs},k}$ is set per leaf type to the standard deviation of the residuals of the direct estimator

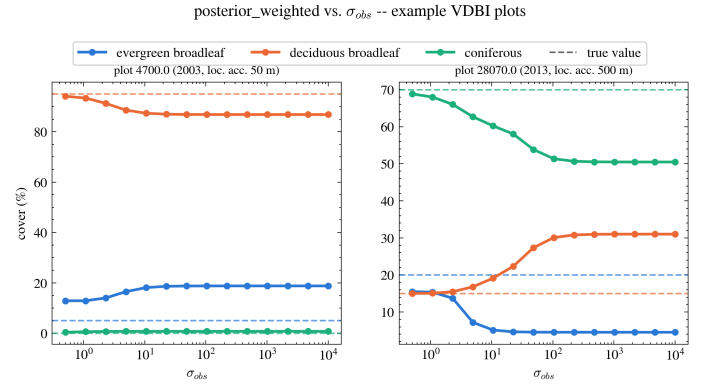


Figure A1. Sensitivity study of the observation bandwidth $\sigma_{\text{obs},k}$ that controls the reference-based likelihood landscape for the posterior-weighted estimator.

on the VPO split, i.e. on the dataset with the lowest location uncertainty, so that the likelihood is not sharper than the residual scatter that is already explained by model error alone. It acts as a single dial between the two limiting cases: $\sigma_{\text{obs},k} \rightarrow \infty$ removes the label from the weighting and recovers a purely prior-weighted neighbourhood mean, whereas $\sigma_{\text{obs},k} \rightarrow 0$ recovers the best-match estimator of Sect. A2. This is similar to a sampling-importance-resampling implementation that draws n candidates directly from the prior and weights them by the likelihood alone (Lobo et al., 2020).

For the DTS task the same weights are applied to the predicted class-probability vector $\hat{\mathbf{p}}(s, t)$. The likelihood of a hypothesis is the probability that the candidate cell assigns to the reference class.

Spatial uncertainty σ_s is set for each plot individually, depicting the reported accuracy of the plot coordinates based on GNSS measurements or data curator expert knowledge. σ_s ranges between 1 m and 1000 m in the VDBI dataset. The sensitivity of the posterior-weighted estimator to the choice of $\sigma_{\text{obs},k}$ is illustrated in Fig. A1, and example posterior landscapes are shown in Fig. A2.

add section to explain the two figures

A2 Best-match estimator

The best-match estimator replaces the weighted mean by a hard selection over a neighbourhood ($2\sigma_s$, plot location accuracy in m, and a fixed window of $t_p \pm 1$ a). For the LTC task it returns, per leaf type, the value of the single candidate cell that agrees most closely with the reference,

$$\hat{y}_{p,k}^{\text{BM}} = \hat{y}_k\left(\arg \min_{(s,t) \in \mathcal{N}_p} |\hat{y}_k(s, t) - y_{p,k}|\right), \quad (\text{A4})$$

and for the DTS task the label of the candidate cell that assigns the highest probability to the reference class. Because the selection is made using the reference value itself, this

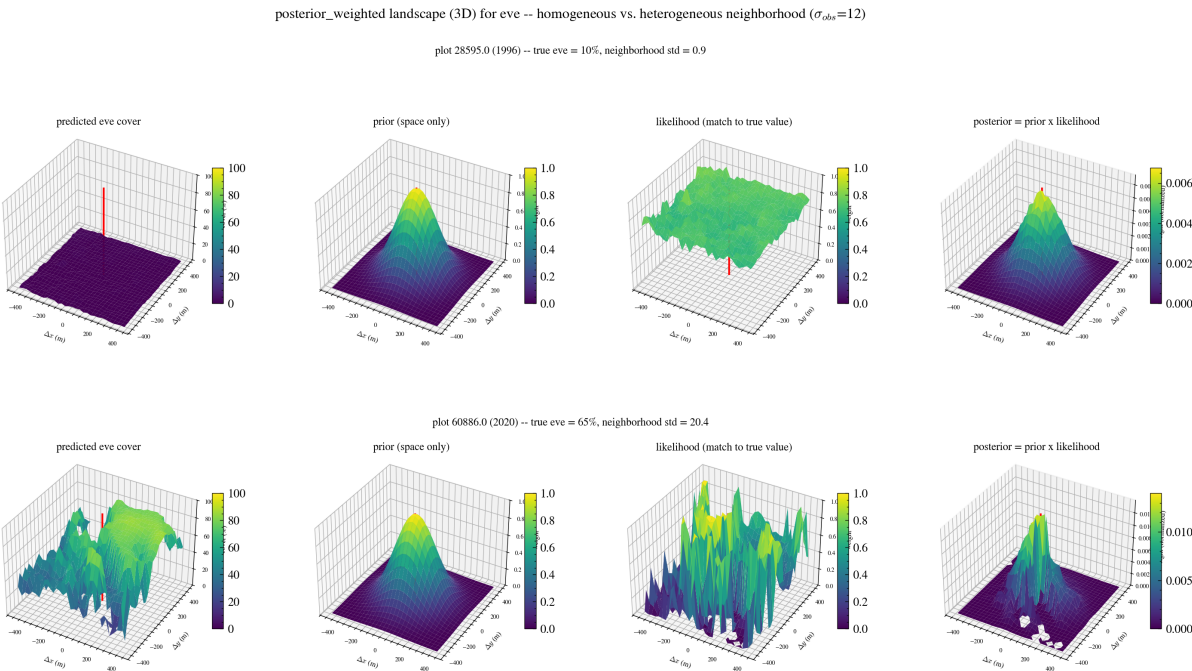


Figure A2. Posterior weight landscape for an example plot with highly heterogenous EVE cover within the neighbourhood computed from the gaussian prior and likelihood for the posterior-weighted estimator.

estimator is optimistically biased by construction: any leaf-type value present anywhere in the neighbourhood can be matched, and a plot is counted as correctly classified whenever at least one candidate cell within the plot location accuracy reported by the data providers and $\pm 1a$ carries the reference class. The resulting metrics are therefore not corrected accuracies but an upper optimistic bound on what could be attributed to spatial(-temporal) reference mismatch rather than to model error.

- figure: sensitivity study for sigmaobs in the posterior method. lower bound direct, upper bound best match, line in between with sigma
- figure of posterior landscape for 2 example plots. prior (gaussian) * likelihood => posterior

Appendix B: Comparison to historical orthophoto imagery

TODO.

Author contributions.

TODO.

Competing interests.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Disclaimer.

TODO.

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References

Alessi, N., Těšitel, J., Zerbe, S., Spada, F., Agrillo, E., and Wellstein, C.: Ancient refugia and present-day habitat suitability of native laurophylls in Italy, *Journal of Vegetation Science*, 30, 564–574, <https://doi.org/10.1111/jvs.12743>, 2019.

Ball, J. G., Wicklein, J. A., Feng, Z., Knezevic, J., Jaffer, S., Madhavapeddy, A., Atzberger, C., Dalponte, M.,

- and Coomes, D.: Geospatial foundation models enable data-efficient tree species mapping in temperate mountain forests, <https://doi.org/10.64898/2026.02.23.707022>, iSSN: 2692-8205 Pages: 2026.02.23.707022 Section: New Results, 2026.
- 5 Bayle, A., Gascoin, S., Berner, L. T., and Choler, P.: Landsat-based greening trends in alpine ecosystems are inflated by multidecadal increases in summer observations, *Ecography*, 2024, e07394, <https://doi.org/10.1111/ecog.07394>, eprint: <https://nsojournals.onlinelibrary.wiley.com/doi/pdf/10.1111/ecog.07394>, Springer Tracts in Civil Engineering, Springer International Publishing, Cham, ISBN 978-3-030-98677-3 978-3-030-98678-0, <https://doi.org/10.1007/978-3-030-98678-0>, 2022.
- 10 Bell, D. M., Gregory, M. J., and Yang, Z.: Hindcasting and updating Landsat-based forest structure mapping across years to support forest management and planning, *Forest Ecology and Management*, 572, 122 239, <https://doi.org/10.1016/j.foreco.2024.122239>, 2024.
- 15 Blickensdörfer, L., Oehmichen, K., Pflugmacher, D., Kleinschmit, B., and Hostert, P.: National tree species mapping using Sentinel-1/2 time series and German National Forest Inventory data, *Remote Sensing of Environment*, 304, 114 069, <https://doi.org/10.1016/j.rse.2024.114069>, 2024.
- 20 Bolyn, C., Lejeune, P., Michez, A., and Latte, N.: Mapping tree species proportions from satellite imagery using spectral-spatial deep learning, *Remote Sensing of Environment*, 280, 113 205, <https://doi.org/10.1016/j.rse.2022.113205>, 2022.
- 25 Bricca, A., Zerbe, S., Sabatini, F. M., Hiebl, B., Rutzinger, M., di Musciano, M., Calvia, G., Chiarucci, A., Poschlod, P., Rossi, C., Alessi, N., and Bonari, G.: Topography and Soil Moisture Regulate the Temperature-Biodiversity Relationship of Forests, *Global Ecology and Biogeography*, 35, e70 186, <https://doi.org/10.1111/geb.70186>, eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/geb.70186>, 2026.
- 30 Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., Shelhamer, E., Lahera, E., Wiles, O., Ilyushchenko, S., Gorelick, N., Zhang, L. L., Alj, S., Schechter, E., Askay, S., Guinan, O., Moore, R., Boukouvalas, A., and Kohli, P.: AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data, <https://doi.org/10.48550/arXiv.2507.22291>, arXiv:2507.22291 [cs], 2025.
- 40 Chastain, R. A. and Townsend, P. A.: Use of Landsat ETM and topographic data to characterize evergreen understory communities in Appalachian deciduous forests, *Photogrammetric Engineering & Remote Sensing*, 73, 563–575, 2007.
- 45 Chelli, S., Wellstein, C., Campetella, G., Canullo, R., Tonin, R., Zerbe, S., and Gerdol, R.: Climate change response of vegetation across climatic zones in Italy, *Climate Research*, 71, 249–262, <https://doi.org/10.3354/cr01443>, 2017.
- Chen, T., Kornblith, S., Norouzi, M., and Hinton, G.: A Simple Framework for Contrastive Learning of Visual Representations, <https://doi.org/10.48550/arXiv.2002.05709>, arXiv:2002.05709 [cs], 2020.
- 50 Conedera, M., Wohlgemuth, T., Tanadini, M., and Pezzatti, G. B.: Drivers of broadleaved evergreen species spread into deciduous forests in the southern Swiss Alps, *Reg Environ Change*, 18, 425–436, <https://doi.org/10.1007/s10113-017-1212-7>, 2018.
- 55 Feng, Z., Atzberger, C., Jaffer, S., Knezevic, J., Sormunen, S., Young, R., Lisaius, M. C., Immitzer, M., Jackson, T., Ball, J., Coomes, D. A., Madhavapeddy, A., Blake, A., and Keshav, S.: TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis, <https://doi.org/10.48550/arXiv.2506.20380>, arXiv:2506.20380 [cs.LG], 2026.
- 60 Gasparini, P., Di Cosmo, L., Floris, A., and De Laurentis, D., eds.: Italian National Forest Inventory—Methods and Results of the Third Survey: Inventario Nazionale delle Foreste e dei Serbatoi Forestali di Carbonio—Metodi e Risultati della Terza Indagine, Springer Tracts in Civil Engineering, Springer International Publishing, Cham, ISBN 978-3-030-98677-3 978-3-030-98678-0, <https://doi.org/10.1007/978-3-030-98678-0>, 2022.
- 65 Grabska-Szwagrzyk, E., Tiede, D., Sudmanns, M., and Kozak, J.: Map of forest tree species for Poland based on Sentinel-2 data, *Earth System Science Data*, 16, 2877–2891, <https://doi.org/10.5194/essd-16-2877-2024>, 2024.
- 70 Grünig, M., Rammer, W., Senf, C., Albrich, K., André, F., Augustynczyk, A. L. D., Baumann, M., Bohn, F. J., Bouwman, M., Bugmann, H., Collalti, A., Cristal, I., Dalmonech, D., De Coligny, F., Dobor, L., Dollinger, C., Espelta, J. M., Forrester, D. I., Garcia-Gonzalo, J., González-Olabarria, J. R., Hiltner, U., Hlásny, T., Honkaniemi, J., Huber, N., Jonard, M., Jönsson, A. M., Kunstler, G., Lagergren, F., Lindner, M., Mina, M., Moos, C., Morin, X., Muys, B., Nabuurs, G.-J., Nieberg, M., Patacca, M., Peltoniemi, M., Reyer, C. P. O., Schelhaas, M.-J., Storms, I., Thom, D., Toïgo, M., and Seidl, R.: Climate change will increase forest disturbances in Europe throughout the 21st century, *Science*, 391, eadx6329, <https://doi.org/10.1126/science.adx6329>, 2026.
- 75 Hemmerling, J., Pflugmacher, D., and Hostert, P.: Mapping temperate forest tree species using dense Sentinel-2 time series, *Remote Sensing of Environment*, 267, 112 743, <https://doi.org/10.1016/j.rse.2021.112743>, 2021.
- 80 Hendrycks, D. and Gimpel, K.: Gaussian Error Linear Units (GELUs), <https://arxiv.org/abs/1606.08415v5>, 2016.
- Herraz, A. D., Salazar-Zarzosa, P., Acosta-Muñoz, C., Hernández-Clemente, R., and Villar, R.: Aridity-induced phenological shifts and greening trends in Mediterranean forest species: Insights from 28 years of Landsat data in southern Spain, *Ecological Indicators*, 171, 113 115, <https://doi.org/10.1016/j.ecolind.2025.113115>, 2025.
- 85 Hiebl, B., Alessi, N., Calvia, G., Bricca, A., Bonari, G., Zangari, G., Dorigo, W., Zerbe, S., and Rutzinger, M.: Advancing forest mapping: Pretraining strategies and deep-ensemble based uncertainty for predicting evergreen broad-leaved cover from Sentinel-2 time series, *International Journal of Applied Earth Observation and Geoinformation*, 142, 104 734, <https://doi.org/10.1016/j.jag.2025.104734>, 2025.
- 90 Hiebl, B., Alessi, N., Calvia, G., Bricca, A., Bonari, G., Zangari, G., Zerbe, S., and Rutzinger, M.: Combining specialized Sentinel-2 time series features with AlphaEarth Foundations for forest type mapping, *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XI-3-2026, 117–124, <https://doi.org/10.5194/isprs-annals-XI-3-2026-117-2026>, conference Name: XXV ISPRS Congress 2026 “From Imagery to Understanding”, Commission III - 4–11 July 2026, Toronto, Canada, 2026.
- 95 Kang, Y., Ozdogan, M., Gao, F., Anderson, M. C., White, W. A., Yang, Y., Yang, Y., and Erickson, T. A.: A data-driven approach to estimate leaf area index for Landsat images over the

- contiguous US, *Remote Sensing of Environment*, 258, 112 383, <https://doi.org/10.1016/j.rse.2021.112383>, 2021.
- Kollert, A., Bremer, M., Löw, M., and Rutzing, M.: Exploring the potential of land surface phenology and seasonal cloud free composites of one year of Sentinel-2 imagery for tree species mapping in a mountainous region, *International Journal of Applied Earth Observation and Geoinformation*, 94, 102 208, <https://doi.org/10.1016/j.jag.2020.102208>, 2021.
- Leonelli, G., Pelfini, M., Morra di Cella, U., and Garavaglia, V.: Climate Warming and the Recent Treeline Shift in the European Alps: the Role of Geomorphological Factors in High-Altitude Sites, *AMBIO*, 40, 264–273, <https://doi.org/10.1007/s13280-010-0096-2>, 2011.
- Lobo, V. G. R., Fonseca, T. C. O., and Moura, F. A. S.: Bayesian cross-validation of geostatistical models, *Spatial Statistics*, 35, 100 394, <https://doi.org/10.1016/j.spasta.2019.100394>, 2020.
- Mañas, O., Lacoste, A., Giró-i Nieto, X., Vazquez, D., and Rodríguez, P.: Seasonal Contrast: Unsupervised Pre-Training from Uncurated Remote Sensing Data, in: 2021 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 9394–9403, ISSN 2380-7504, <https://doi.org/10.1109/ICCV48922.2021.00928>, iSSN: 2380-7504, 2021.
- Mattioli, W., R, R., D, B., G, C., G, D., D, G., M, P., and P, C.: La Carta Forestale d'Italia (CFI2020): un ritratto aggiornato dei boschi italiani, *Forest@ - Journal of Silviculture and Forest Ecology*, 22, 39, <https://doi.org/10.3832/efor4836-022>, 2025.
- Midolo, G., Clark, A. T., Chytrý, M., Essl, F., Dullinger, S., Jandt, U., Bruehlheide, H., Dengler, J., Axmanová, I., Ačić, S., Argagnon, O., Biurrun, I., Bonari, G., Chiarucci, A., Čušterevska, R., De Frenne, P., De Sanctis, M., Divíšek, J., Doležal, J., Dziuba, T., Ejrnæs, R., Garbolino, E., Jentsch, A., Jiménez-Alfaro, B., Lenoir, J., Moeslund, J. E., Napoleone, F., Rumpf, S. B., Svenning, J.-C., Swacha, G., Tatarenko, I., Večeřa, M., Vynokurov, D., and Keil, P.: Sixty years of plant community change in Europe indicate a shift toward nutrient-richer and denser vegetation, *Science Advances*, 12, eaeb2493, <https://doi.org/10.1126/sciadv.aeb2493>, 2026.
- Miettinen, J., Breidenbach, J., Adame, P., Adolt, R., Alberdi, I., Antropov, O., Arnarsson, Ó., Astrup, R., Berger, A., Bogason, J., Chirici, G., Corona, P., D'Amico, G., Fejfar, J., Fischer, C., Gohon, F., Gschwantner, T., Hertzler, J., Koma, Z., Korhonen, K. T., Krajnc, L., Latte, N., Lejeune, P., McCullagh, A., Mionskowski, M., Moreno-Fernández, D., Myllymäki, M., Nilsson, M., Perin, J., Pitkänen, J., Redmond, J., Riedel, T., Schumacher, J., Seitsonen, L., Sirro, L., Skudnik, M., Snorrason, A., Sroga, R., Traub, B., Traustason, B., Westerlund, B., and Wurrpillot, S.: Pan-European forest maps produced with a combination of earth observation data and national forest inventory plots, *Data in Brief*, 60, 111 613, <https://doi.org/10.1016/j.dib.2025.111613>, 2025.
- Obuchowicz, C., Poussin, C., and Giuliani, G.: Change in observed long-term greening across Switzerland – evidence from a three decades NDVI time-series and its relationship with climate and land cover factors, *Big Earth Data*, 8, 1–32, <https://doi.org/10.1080/20964471.2023.2268322>, <https://doi.org/10.1080/20964471.2023.2268322>, 2024.
- Olofsson, P., Foody, G. M., Herold, M., Stehman, S. V., Woodcock, C. E., and Wulder, M. A.: Good practices for estimating area and assessing accuracy of land change, *Remote Sensing of Environment*, 148, 42–57, <https://doi.org/10.1016/j.rse.2014.02.015>, 2014.
- Persson, H. J., Ekström, M., and Ståhl, G.: Quantify and account for field reference errors in forest remote sensing studies, *Remote Sensing of Environment*, 283, 113 302, <https://doi.org/10.1016/j.rse.2022.113302>, 2022.
- Pflugmacher, D., Rabe, A., Peters, M., and Hostert, P.: Mapping pan-European land cover using Landsat spectral-temporal metrics and the European LUCAS survey, *Remote Sensing of Environment*, 221, 583–595, <https://doi.org/10.1016/j.rse.2018.12.001>, 2019.
- Rußwurm, M. and Körner, M.: Self-attention for raw optical Satellite Time Series Classification, *ISPRS Journal of Photogrammetry and Remote Sensing*, 169, 421–435, <https://doi.org/10.1016/j.isprsjprs.2020.06.006>, 2020.
- Safonova, A., Ghazaryan, G., Stiller, S., Main-Knorn, M., Nendel, C., and Ryo, M.: Ten deep learning techniques to address small data problems with remote sensing, *International Journal of Applied Earth Observation and Geoinformation*, 125, 103 569, <https://doi.org/10.1016/j.jag.2023.103569>, 2023.
- Shannon, E. S., Finley, A. O., Hayes, D. J., Noralez, S. N., Weiskittel, A. R., Cook, B. D., and Babcock, C.: Quantifying and correcting geolocation error in spaceborne LiDAR forest canopy observations using high spatial accuracy data: A Bayesian model approach, *Environmetrics*, 35, e2840, <https://doi.org/10.1002/env.2840>, <https://onlinelibrary.wiley.com/doi/pdf/10.1002/env.2840>, 2024.
- Strunk, J. L., Reutebuch, S. E., McGaughey, R. J., and Andersen, H.-E.: An examination of GNSS positioning under dense conifer forest canopy in the Pacific Northwest, USA, *Remote Sensing Applications: Society and Environment*, 37, 101 428, <https://doi.org/10.1016/j.rsase.2024.101428>, 2025.
- Sylvain, J.-D., Drolet, G., Thiffault, É., and Anctil, F.: High-resolution mapping of tree species and associated uncertainty by combining aerial remote sensing data and convolutional neural networks ensemble, *International Journal of Applied Earth Observation and Geoinformation*, 131, 103 960, <https://doi.org/10.1016/j.jag.2024.103960>, 2024.
- Tan, J., Li, J., Ma, T., Yan, X., and Huo, Z.: Leveraging Sentinel-1/2 time series and deep learning for accurate forest tree species mapping, *Frontiers in Forests and Global Change*, 8, <https://doi.org/10.3389/ffgc.2025.1599510>, 2025.
- Tong, X., Brandt, M., Yue, Y., Zhang, X., Fensholt, R., Ciais, P., Wang, K., Liu, S., Zhang, W., Mao, C., and Jepsen, M. R.: Reforestation policies around 2000 in southern China led to forest densification and expansion in the 2010s, *Communications Earth & Environment*, 4, 260, <https://doi.org/10.1038/s43247-023-00923-1>, 2023.
- Turbanova, S., Potapov, P., Hansen, M. C., Li, X., Tyukavina, A., Pickens, A. H., Hernandez-Serna, A., Arranz, A. P., Guerra-Hernandez, J., Senf, C., Häme, T., Valbuena, R., Eklundh, L., Brovkina, O., Navrátilová, B., Novotný, J., Harris, N., and Stolle, F.: Tree canopy extent and height change in Europe, 2001–2021, quantified using Landsat data archive, *Remote Sensing of Environment*, 298, 113 797, <https://doi.org/10.1016/j.rse.2023.113797>, 2023.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I.: Attention

- Is All You Need, <https://doi.org/10.48550/arXiv.1706.03762>, arXiv:1706.03762 [cs], 2023.
- Viana-Soto, A. and Senf, C.: The European Forest Disturbance Atlas: a forest disturbance monitoring system using the Landsat archive, *Earth System Science Data*, 17, 2373–2404, <https://doi.org/10.5194/essd-17-2373-2025>, 2025.
- Wang, M., Zheng, Y., Huang, C., Meng, R., Pang, Y., Jia, W., Zhou, J., Huang, Z., Fang, L., and Zhao, F.: Assessing Landsat-8 and Sentinel-2 spectral-temporal features for mapping tree species of northern plantation forests in Heilongjiang Province, China, *Forest Ecosystems*, 9, 100 032, <https://doi.org/10.1016/j.fecs.2022.100032>, 2022.
- Wulder, M. A., Roy, D. P., Radeloff, V. C., Loveland, T. R., Anderson, M. C., Johnson, D. M., Healey, S., Zhu, Z., Scambos, T. A., Pahlevan, N., Hansen, M., Gorelick, N., Crawford, C. J., Masek, J. G., Hermosilla, T., White, J. C., Belward, A. S., Schaaf, C., Woodcock, C. E., Huntington, J. L., Lymburner, L., Hostert, P., Gao, F., Lyapustin, A., Pekel, J.-F., Strobl, P., and Cook, B. D.: Fifty years of Landsat science and impacts, *Remote Sensing of Environment*, 280, 113 195, <https://doi.org/10.1016/j.rse.2022.113195>, 2022.
- Yel, S. G., Özmen, H. B., Öney Birol, S., Tunç Görmüş, E., and Kaplan, G.: Remote sensing applications for assessing climate change impacts on deciduous forests—A systematic review, *Physics and Chemistry of the Earth, Parts A/B/C*, 143, 104 321, <https://doi.org/10.1016/j.pce.2026.104321>, 2026.
- Yuan, Y., Lin, L., Liu, Q., Hang, R., and Zhou, Z.-G.: SITS-Former: A pre-trained spatio-spectral-temporal representation model for Sentinel-2 time series classification, *International Journal of Applied Earth Observation and Geoinformation*, 106, 102 651, <https://doi.org/10.1016/j.jag.2021.102651>, 2022.
- Yuan, Y., Lin, L., Xin, Q., Zhou, Z.-G., and Liu, Q.: An Empirical Study on Data Augmentation for Pixelwise Satellite Image Time-Series Classification and Cross-Year Adaptation, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18, 5172–5188, <https://doi.org/10.1109/JSTARS.2025.3527017>, 2025.
- Yuan, Y., Lin, L., and Lei Lin: Self-Supervised Pretraining of Transformers for Satellite Image Time Series Classification, 14, 474–487, <https://doi.org/10.36227/techrxiv.13025039.v2>, 2020.
- Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., and Arino, O.: ESA WorldCover 10 m 2021 v200, <https://doi.org/10.5281/zenodo.7254221>, 2022.
- Zerveas, G., Jayaraman, S., Patel, D., Bhamidipaty, A., and Eickhoff, C.: A Transformer-based Framework for Multivariate Time Series Representation Learning, <https://doi.org/10.48550/arXiv.2010.02803>, 2020.